

HC Remit Proposal Document

5th January,2018

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1. Introduction

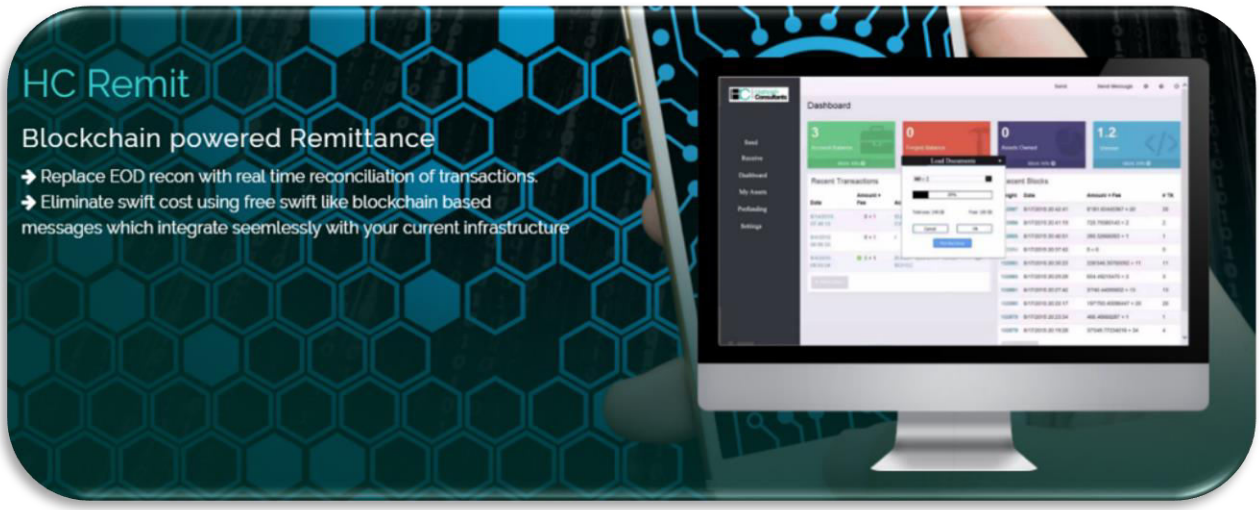
The following proposal is for the implementation of the HC Remit product for a real-time settlement for remittances using blockchain. The scope of this document is limited to a pilot phase and production implementation between Bank's and a Remittance Partner.

The proposed implementation leverages a blockchain platform to exchange remittance messages real time between banks and remittance exchanges. Settlement is carried out by debiting a prefunded account maintained by the receiving bank. Invoices and AML/KYC information can also be exchanged securely by recording the bytecode over blockchain and sharing the document using No SQL databases.

The implementation is expected to significantly reduce transaction costs and increase efficiency owing toinstant reconciliation, visibility of prefunded account and elimination of Swift messaging costs.

2. HC Remit Product description

HC Remit is Hashcash's product for faster low cost cross currency payments.



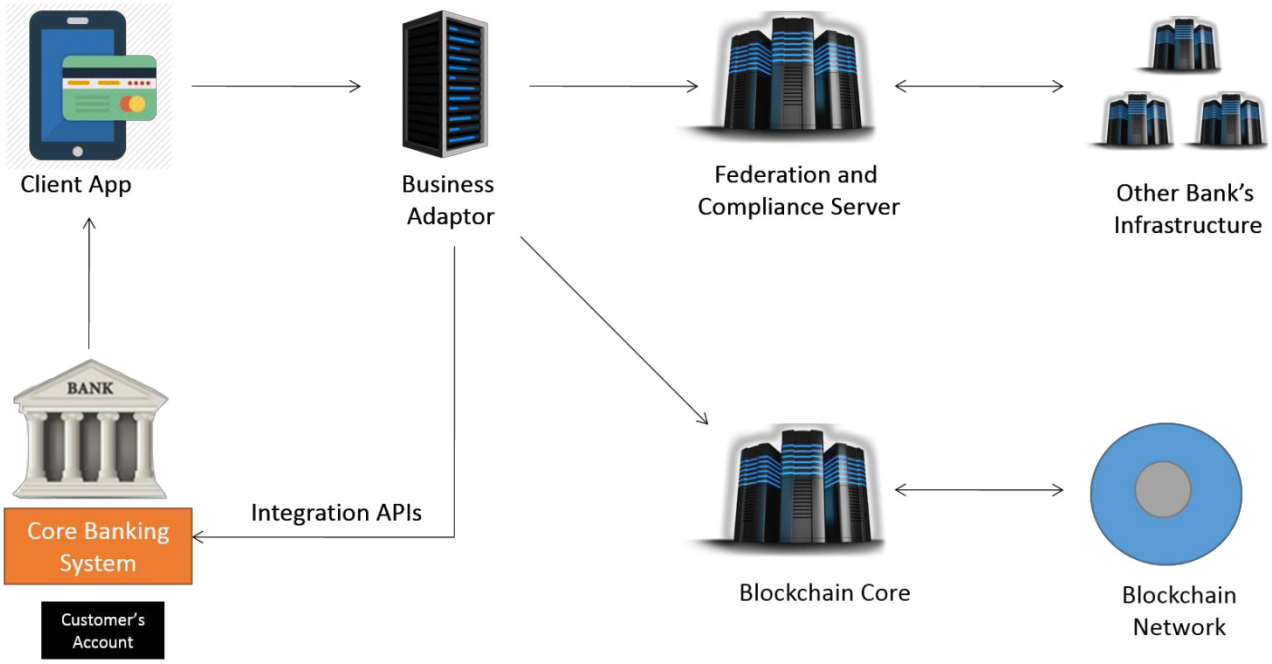
Product Benefits

- Eliminate higher transaction cost arising from using SWIFT networks and Payment Gateways.
- Onboard new banks to your cross-border network faster as opposed to traditional remittance systems.
- Automate critical business processes using Smart Contracts.
- Lower maintenance and recurring costs for your remittance system.
- Safe, secure and authentic transaction handling.

Product Features

- Live tracking of transactions with all parties using the Remittance Dashboard.
- Monitor your funds and NOSTRO accounts through real time tracking.
- Cross currency Remittance.
- New banks can be easily on boarded to the network.

High Level Architecture



3. International Remittance inBeneficiary Country

The below use case enables customers registered with Bank to remit to Beneficiary country. Settlement is real time and messaging on SWIFT system is replaced by API services and blockchain transactions.

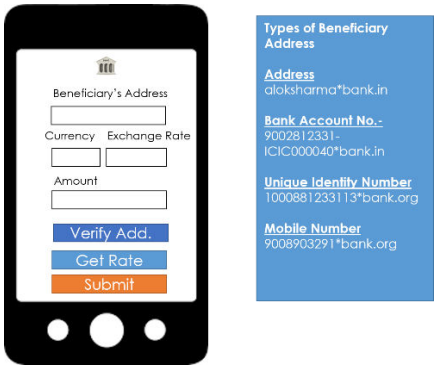
Customer Flow

1. Customer registers with an account with a Bank and is assigned a friendly ID such as Bob@alrajhi.com

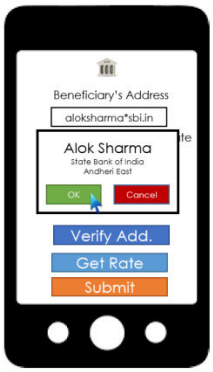
2. He goes to the retail remittance page and is shown a form as below:



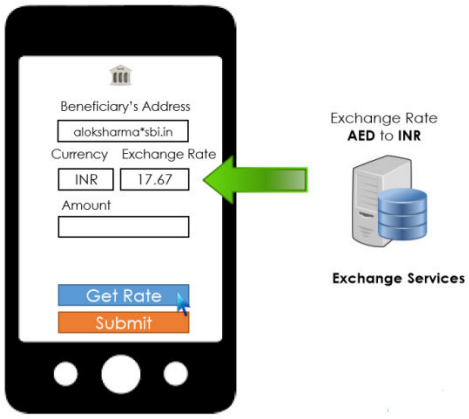
4. He enter the beneficiary address for the transaction. Instead of Bank Account No., Beneficiary Name, Routing Number, SWIFT Code, IFSC Code etc., the proposed system requires that user only provides the receiver's ID that is provided to him when signing up. This ID is issued by the bank or can be asked from the user through a registration process. An existing identifier such as Bank Account No., Unique Identification Number, Mobile number can be used.



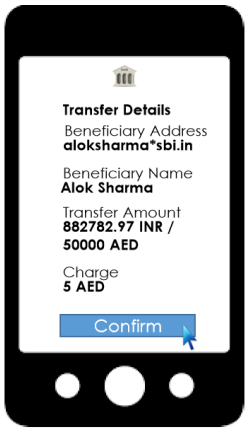
5. The beneficiary address that is submitted can also be verified before the actual transaction is submitted for remittance. This helps avoid reworking on the customer and bank's part if the transaction details are incorrect. The customer is shown the beneficiary details before submission and the transaction only goes through once the details are verified.



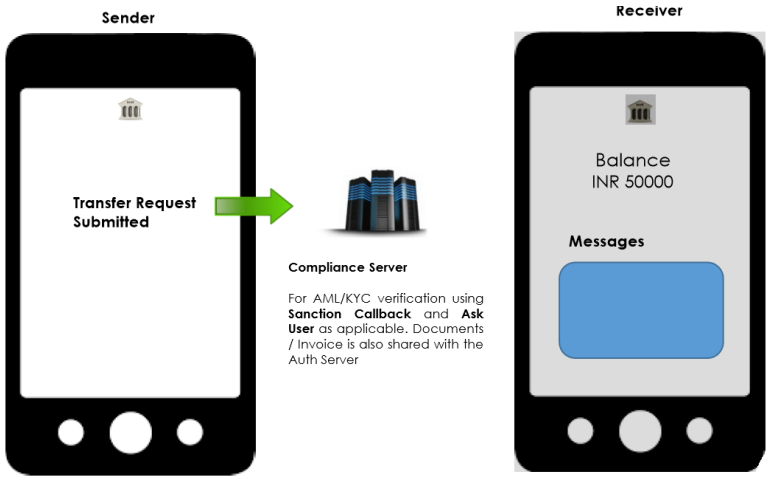
6. The customer enters the currency to be remitted and the amount to be remitted. He is shown the current exchange rate from the bank's APIs or from Global Distributed Exchange's orderbook if so desired.



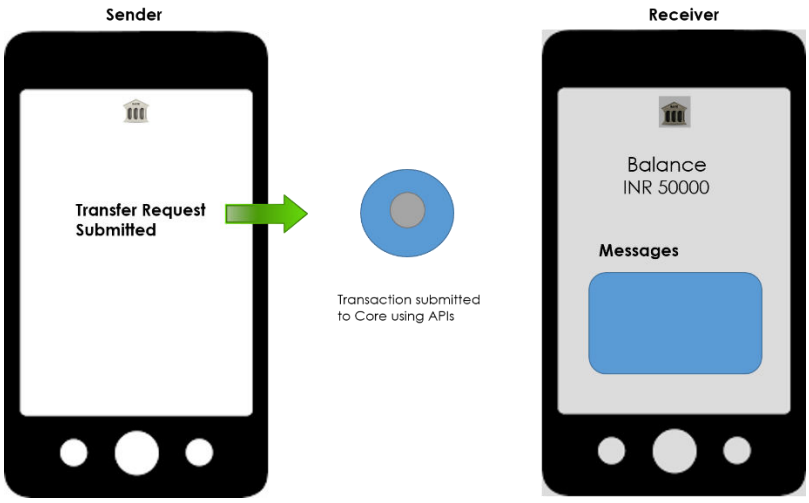
7. The customer is shown the amount that will be credited to the beneficiary bank account and the fees that he needs to pay for the remittance. He needs to acknowledge the details to proceed with the remittance.



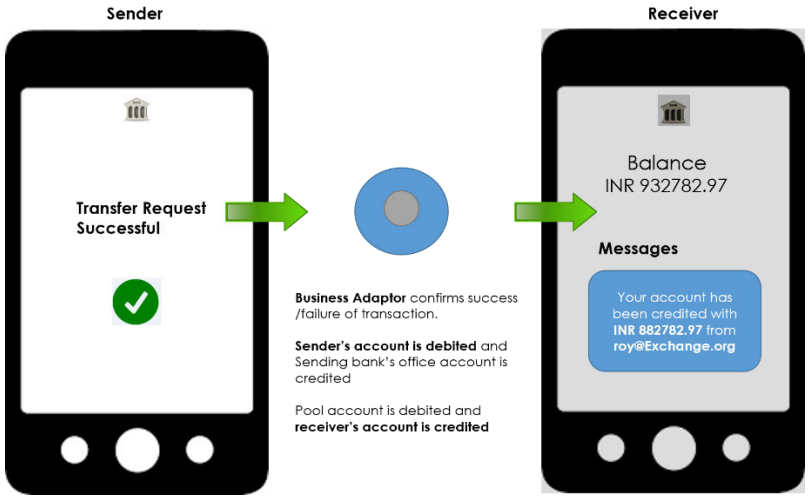
8. Once the customer confirms the details, a compliance check for the Sender and Receiver of the transaction is carried out. The sender's compliance details such as KYC/AML/CTF are verified by the sender's compliance server. The remittance initiating party then shares the details of the transaction with the receiving party. The receiving party's compliance server then does a complete compliance check for the beneficiary's details. If all the compliance requisites are met, both the parties agree to initiate the transaction.



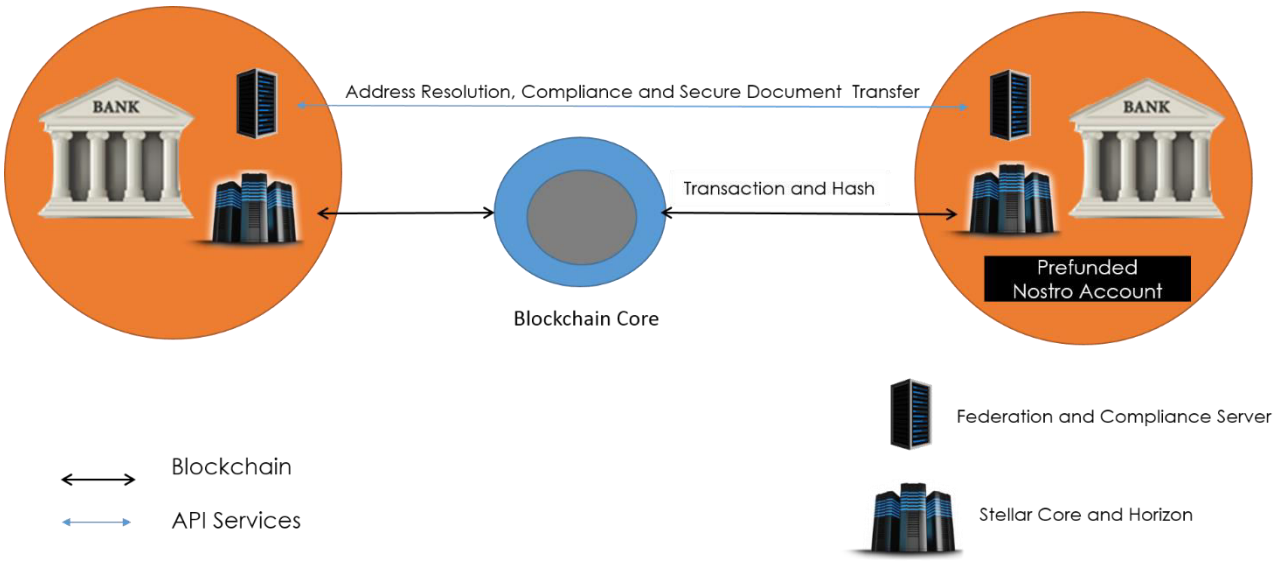
9. Once compliance checks are completed the transaction is submitted to blockchain.



10. The receiving party receives the transaction on the blockchain. It verifies with the compliance server that the incoming transaction has been authorized and all compliance requirements have been met. Once these details are verified, the receiving end debits the office account and credits the beneficiary's account.

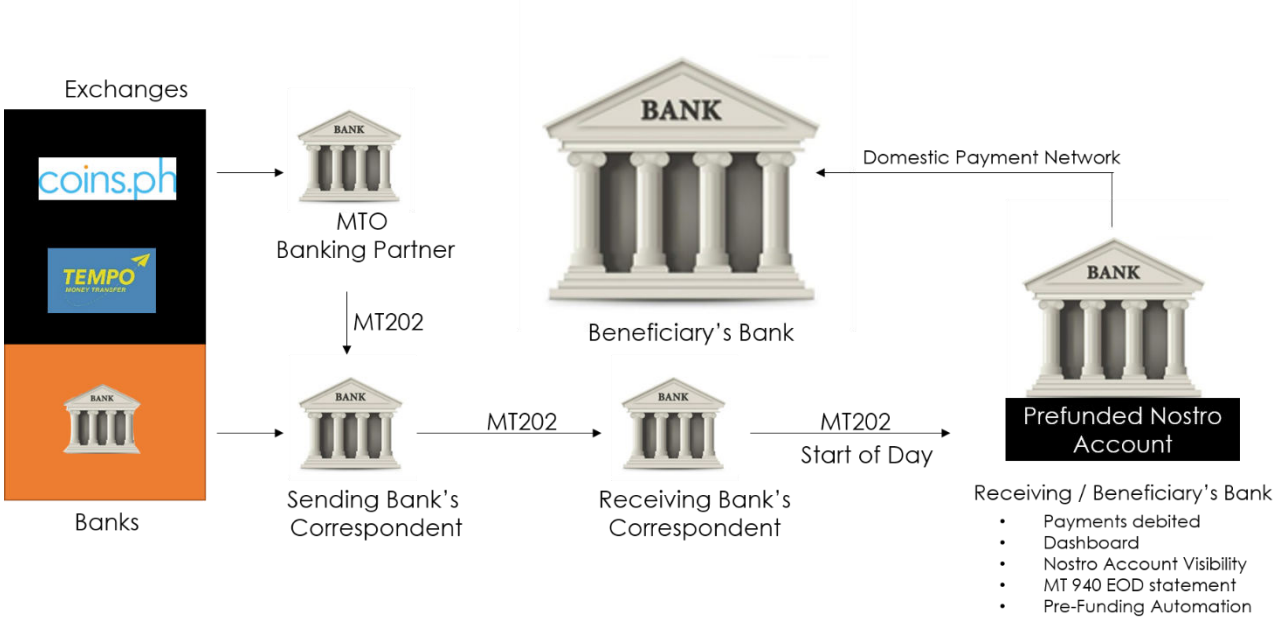


Proposed Model

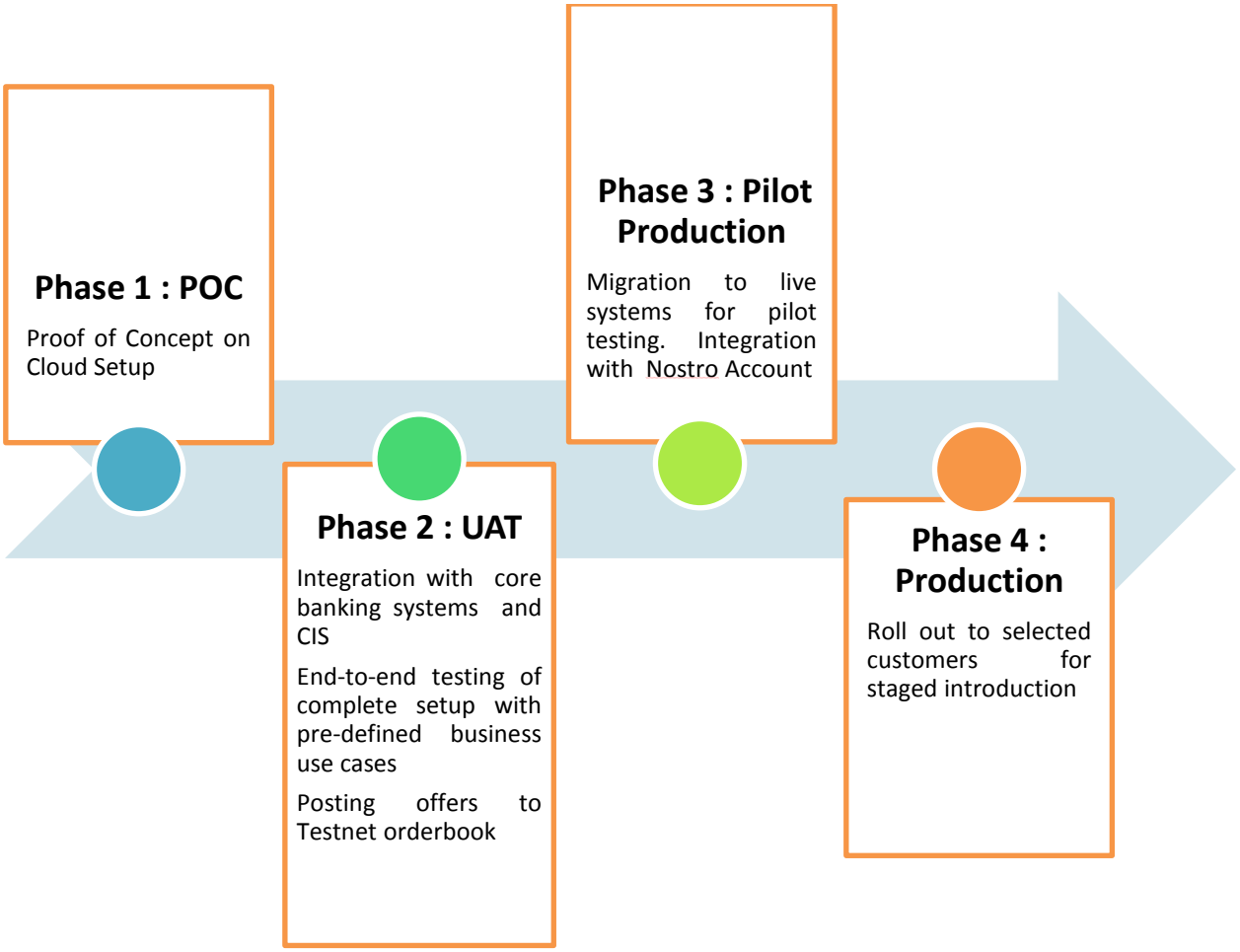


Movement of Fiat Currency

Actual Settlement between the transacting parties happens through established conventional means. the sending party holds pre-funded Nostro accounts either through direct or correspondent relationship with the receiving party. Alternatively, transacting parties can net off transactions and then settle either at the end of the day or any other designated time period. Transaction to domestic banks not on network can be initiated through the domestic payment network in Beneficiary Country (Money Transfer Solutions).



Proposal for Project



POC Scope:

- POC Web-appwith Business Adaptor integrated with network
- Compliance and Federation servers integrated with test databases
- End-to-end transaction on Blockchain with Test Nodes

Pilot Scope:

- Client-side app for UAT testing
- Integration with Payment Processor, regulatory systems and compliance databases

- Report generation and system monitoring screen

Production Scope:

- Client App and APIs for remittance to beneficiary's
- Real time settlement
- Dashboard with visibility of Prefunded account
- Monitoring tools for both parties

Proposed Technology:

- HC Net
- Oracle 12C database for Account Management and Audit
- Java for Front End development and APIs for integration
- Apache Tomcat Webserver
- RHEL 7.0 Operating System for Servers
- Amazon AWS for POC and hosting Node

Project Benefits

- Remittances to other countries using UIN's / Mobile Number / UPI VPAs
- Faster Reconciliation and KYC and AML verification
- Elimination of SWIFT costs
- Generation and Transfer of Nostro/Prefunded Account Statement
- Data stored on blockchain for auditory purpose
- Secure document transfer
- Operational cost reduction

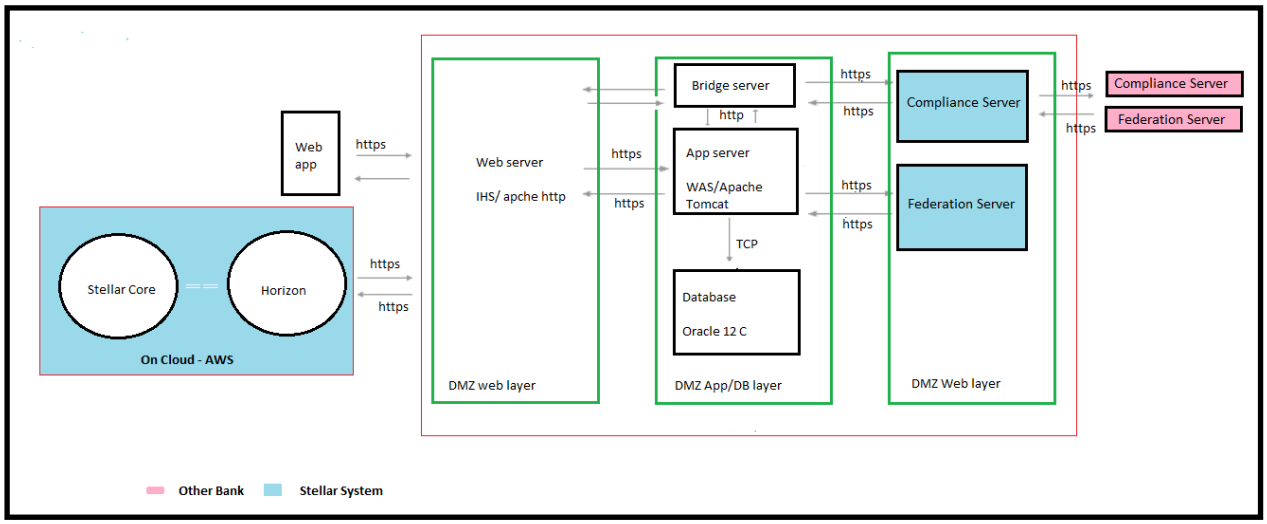
Platform Benefits

- Greater transaction per second on network - 3000 TPS and 290 million transactions per day
- Low cost per transaction and Open Source Platform
- Customer data is anonymous on blockchain
- Proven scalability as operates a global public network

Applications

- Retail Remittances
- International E-commerce payments
- Payment disbursements

4. Proposed Architecture

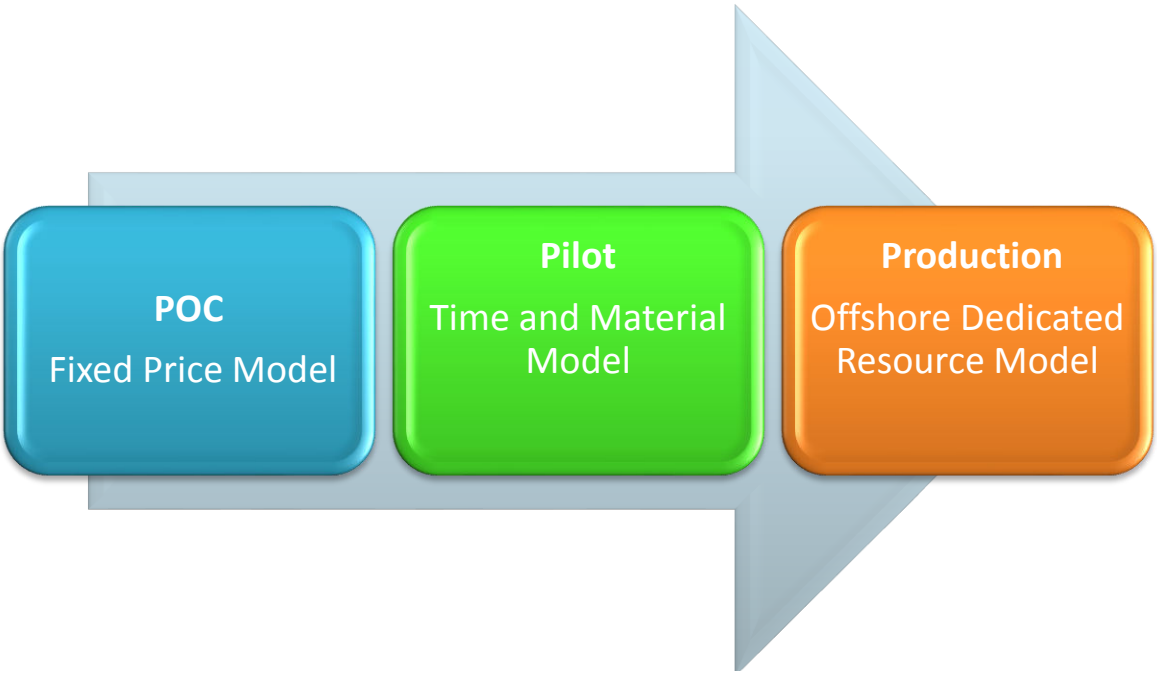


1. 3-point architecture with separation of Web, App and DB servers

2. All rest services communication protocol are https

3 Information shared over the cloud setup does not contain any customer data

5. Engagement Model



6. Commercial Terms

Kindly find the proposed commercials for the project below. The below commercials are as per the scope of the project as per our discussions till now.

Phase	Phase Description	Bill amount to be cleared
		Offsite Consulting Model
R0	• POC - Exhibit HC Remit features in POC	0 USD
R1	• Requirement Gathering • Core and Horizon Server Setup • Compliance server setup • Federation Server Setup • Bank Dashboard • Unit Testing and System Testing with Dummy Data	12500 USD on completion of 80% of ST Test Cases
R2	• Integration with core banking system • KYC/AML system integration • Nostro Visibility • Report Generation • Client App Integration • UT and ST with Dummy Data	7500 USD on completion of 80% of ST Test Cases
R3	• UAT on live customer data replica	5000 USD on completion*
R4	• AMC /production cost in post-production phase - First Year	15000 USD** for production support, application maintenance, customization for other business use cases, public network transaction cost and infrastructure maintenance
R5	• AMC /production cost in post-production phase - Second Year	15000USD*** for production support, application maintenance, customization for additional partners and infrastructure maintenance

*Total implementation fee is USD 25000, last instalment of USD 5000 to be paid on completion of Go-live
**Fee for transacting in Network for first year, to be paid from the day of completion of one year from the day of completion of UAT
*** Fee for transacting in Network for second year, to be paid from the day of completion of two years from the day of completion of UAT

AMC Scope

- Customization of HC Remit for additional Partners

- Annual Maintenance of Platform and Integration
- Production Support
- Infrastructure Maintenance
- Transaction Costs

Terms and Conditions

- The commercials defined are limited to the scope defined in the proposal.
- Any requirement gap found after requirement gathering freezes will be charged separately based on the execution phase of the project, timeline & volume of change as CR implementation phase.
- Please note any requirement change at the time of UAT would cause the CR costing higher as it may cause to UAT regression testing.
- All commercials quoted in this proposal are exclusive of any in-country and withholding taxes. These will be applicable and payable by the customer.
- Hashcash Consultants shall not be responsible for any installation, commissioning or maintenance of any software or hardware unless supplied by Hashcash Consultants
- Customer to be solely responsible for the content of the data and materials transmitted on the services and blockchain.